PORT SCAN DETECTION USING SPLUNK SIEM

1. ABSTRACT

In modern cybersecurity environments, detecting malicious activities such as port scanning is critical for protecting network infrastructure. This project presents the design and implementation of a Security Information and Event Management (SIEM) system using Splunk to identify suspicious behavior through log analysis.

A virtual lab environment was created using Ubuntu as the Splunk server and Kali Linux as the attacker machine. System logs were ingested into Splunk and analyzed using custom queries. Since raw logs lacked structured network fields, anomaly-based detection was implemented by identifying unusual spikes in log activity.

The system successfully detects abnormal patterns that indicate potential reconnaissance activities, demonstrating how SIEM tools can be effectively used in real-world Security Operations Center (SOC) environments.

2. INTRODUCTION

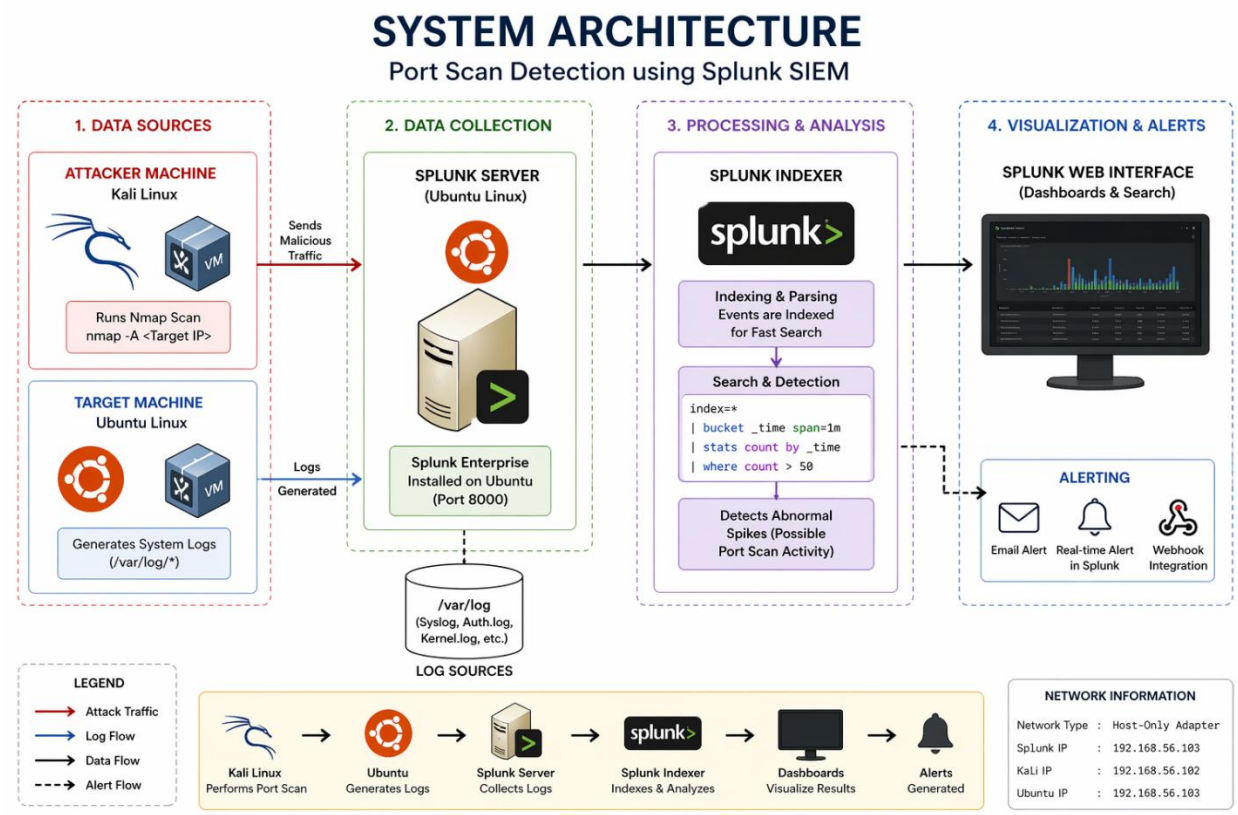
With the rapid growth of cyber threats, organizations rely on SIEM systems to monitor, detect, and respond to security incidents. Port scanning is a common reconnaissance technique used by attackers to identify open ports and vulnerabilities in a system.

Splunk is a widely used SIEM platform that enables real-time log collection, analysis, and visualization. This project focuses on leveraging Splunk to detect port scanning activity by analyzing Linux system logs and identifying anomalies in event patterns.

3. OBJECTIVES

- To understand the working of SIEM tools (Splunk)
- To collect and analyze system logs from a Linux environment
- To simulate network scanning attacks using Kali Linux
- To implement anomaly-based detection techniques
- To generate alerts for suspicious activity
- To visualize attack patterns using dashboards

4. SYSTEM ARCHITECTURE



Architecture Description

The system consists of three main components:

1. **Attacker Machine (Kali Linux)**
   - Performs reconnaissance using tools like Nmap
   - Generates suspicious traffic and system events
2. **Target & SIEM Server (Ubuntu with Splunk)**
   - Hosts Splunk Enterprise
   - Collects logs from /var/log directory
   - Processes and indexes data
3. **Splunk Processing Layer**
   - Indexes incoming logs
   - Executes search queries

- Detects anomalies
- Generates alerts and visualizations

5. TOOLS AND TECHNOLOGIES USED

Tool / Technology	Description
Splunk Enterprise	SIEM tool for log analysis and monitoring
Kali Linux	Penetration testing OS for attack simulation
Ubuntu Linux	Log-generating system and Splunk host
VirtualBox	Virtualization platform
Nmap	Network scanning tool

6. METHODOLOGY

6.1 Environment Setup

- Installed Ubuntu and Kali Linux on VirtualBox
- Configured both machines on the same network

6.2 Splunk Installation

- Installed Splunk Enterprise on Ubuntu
- Accessed Splunk Web Interface:

http://<ubuntu-ip>:8000

6.3 Data Ingestion

- Added data source from:

/var/log

- Selected source type:

linux_messages_syslog

6.4 Attack Simulation

Port scanning was performed using Kali Linux:

nmap -A <target-ip>

This generated multiple system log entries in a short time.

6.5 Detection Strategy

Since logs did not contain structured fields like src_ip and dest_port, an **anomaly-based detection approach** was used.

Splunk Query Used:

```
index=*
| bucket _time span=1m
| stats count by _time
| where count > 50
```

Explanation:

- Groups logs into 1-minute intervals
- Counts number of events per minute
- Detects unusually high activity

6.6 Visualization

- Used **Column Chart** in Splunk
- Observed spikes during attack simulation

6.7 Alert Configuration

- Alert Name: *Port Scan Detection*
- Condition: Event count exceeds threshold
- Type: Real-time / Scheduled

7. RESULTS AND ANALYSIS

The system successfully identified abnormal spikes in log activity.

Observations:

- Normal activity → Low event count
- Attack activity → Sudden spike (2000–3000+ events)

These spikes correspond to scanning activity performed using Nmap.

Interpretation:

The detection mechanism effectively identifies suspicious behavior based on log volume anomalies, which is a common approach in real-world SOC environments.

8. LIMITATIONS

- Lack of structured fields such as source IP and destination port
- Detection is anomaly-based rather than signature-based
- No integration with advanced threat intelligence

9. FUTURE ENHANCEMENTS

- Integrate **Splunk Universal Forwarder**
- Use **Metasploitable** for realistic attack scenarios
- Implement field-based detection:

stats dc(dest_port) by src_ip

- Create advanced dashboards
- Enable email or webhook alerts

10. CONCLUSION

This project demonstrates the practical implementation of a SIEM-based detection system using Splunk. By analyzing system logs and applying anomaly detection techniques, it is possible to identify suspicious activities such as port scanning.